Sahiti Bommareddy

AI Agent Platforms · Distributed Systems · Resilient Critical Infrastructure

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EDUCATION

Ph.D., Computer Science, Johns Hopkins University, Baltimore, MD — 2019–2025

Advisor: Yair Amir · Dissertation: *Real-Time Byzantine-Resilient Power Grid Infrastructure*

M.S., Computer Science, Duke University, Durham, NC — 2017–2019

B.Tech., Electronics & Communication Engineering, Jawaharlal Nehru Technological University, Hyderabad, India — 2007–2011

RESEARCH & TECHNICAL PROFILE

Agent product-platform architecture and resilient distributed systems, with emphasis on production AI agents, governed external-action execution, workflow orchestration, Byzantine fault tolerance, intrusion tolerance, real-time systems, cybersecurity, resilient networking, and machine-learning-based intrusion detection for critical infrastructure.

PROFESSIONAL & RESEARCH EXPERIENCE

Ema — Software Engineer — Feb 2025–Present

- Own major areas of Ema's integrations and external-actions platform and designed the agent that turns API specifications, documentation, MCP servers, and other system descriptions into validated, test-run, publishable tools without hand-written integration code.
- Designed Ema's multi-tool planning and execution agent and drove shared agent-harness capabilities spanning planning and replanning, dependency resolution, validation, evaluation, safeguards, and configurable human approval.
- Designed governed tool execution with isolated runtimes, sandboxing, network restrictions, identity boundaries, credential isolation, recovery controls, and per-call auditability.
- Owned event-driven and long-running execution across schedules, webhooks, polling, dynamic workflows, durable state, deduplication, re-drive, and partial-failure recovery.
- Designed MCP support and no-code capability creation; drove connector-catalog lifecycle, authentication, permission, CI/CD, quality, and distribution capabilities.
- Set technical direction through architecture reviews, closely mentored three engineers, and coordinated delivery across more than 15 geographically distributed engineers.
- Developed production observability and structured failure attribution across agent execution, reducing detection and resolution time by more than 70%.

Johns Hopkins University — Doctoral Researcher, Distributed Systems and Networks Lab — 2019–2025

- Designed and built real-time distributed systems for critical-infrastructure protection, including power-grid substations, combining Byzantine fault-tolerant replication, resilient networking, and proactive recovery.
- Developed cybersecurity and machine-learning capabilities for network intrusion detection and situational awareness integrated with resilient control systems.
- Developed methods for analyzing and tolerating compound cyber and physical threats to critical-infrastructure control systems; research received the IEEE SRDS 2024 Best Paper Award.
- Co-created Spire and contributed to Prime and Spines; delivered resilient-system research for national-laboratory and major power-grid-vendor deployment environments.

PEER-REVIEWED PUBLICATIONS

Sahiti Bommareddy, Maher Khan, Huzaifah Nadeem, Benjamin Gilby, Imes Chiu, John W. van de Lindt, Omar Nofal, Mathaios Panteli, Linon Wells II, Yair Amir, and Amy Babay. "Tolerating Compound Threats in Critical Infrastructure Control Systems." *IEEE Symposium on Reliable Distributed Systems (SRDS)*, 2024. **Best Paper Award.**

Christopher Bonebrake, Jonathan Sebastian-Cardenas, Carl H. Miller, **Sahiti Bommareddy**, Yair Amir, Kade Cornelison, Cliff Eyre, Paul Skare, Sri Nikhil Gupta Gourisetti, Aditya Ashok, and Bev Johnson. "ByzSec—A Multi-layered Byzantine Resilient Architecture for Bulk Power System Protective Relays." *IEEE Power & Energy Society General Meeting*, 2024.

Sahiti Bommareddy, Maher Khan, David J. Sebastian Cardenas, Carl Miller, Christopher Bonebrake, Yair Amir, and Amy Babay. "Real-Time Byzantine Resilient Power Grid Infrastructure: Evaluation and Trade-offs." *Workshop on Engineering of Reliable, Secure and Adaptive Systems at IEEE RTSS*, 2022.

Sahiti Bommareddy, Daniel Qian, Christopher Bonebrake, Paul Skare, and Yair Amir. "Real-Time Byzantine Resilience for Power Grid Substations." *IEEE Symposium on Reliable Distributed Systems (SRDS)*, 2022.

Sahiti Bommareddy, Benjamin Gilby, Maher Khan, Imes Chiu, Mathaios Panteli, John W. van de Lindt, Yair Amir, and Amy Babay. "Data-Centric Analysis of Compound Threats to Critical Infrastructure Control Systems." *Workshop on Data-Centric Dependability and Security at IEEE/IFIP DSN*, 2021.

OPEN-SOURCE SYSTEMS

Spire — Co-creator

Open-source intrusion-tolerant SCADA and substation-protection system engineered to withstand system- and network-level compromise while meeting power-grid timing requirements, including 100–200 ms control-center updates and 4.167 ms autonomous substation-protection actions.

- **Spire 1.3:** Optimized transaction latency by approximately 3× and integrated machine-learning-based network intrusion detection.
- **Spire 2.0:** Developed the first real-time Byzantine-resilient protection system for power-grid substations; delivered it for deployment at Pacific Northwest National Laboratory, Siemens, GE, and Hitachi Energy, and supported adversarial red-team evaluation by Sandia National Laboratories.
- **Spire 2.1:** Added reconfiguration capabilities to sustain operation under compound cyber and physical threats.

Prime — Contributor

Intrusion-tolerant replication engine implementing Byzantine fault-tolerant replication with performance guarantees under attack and reconfiguration for enhanced resilience.

Spines — Contributor

Resilient overlay-messaging infrastructure providing automatic reconfiguration and network flexibility for research and production deployments with demanding latency, reliability, and resilience requirements.

TECHNICAL SKILLS

Agent platforms	Agent building, agent harness, multi-tool planning and execution, workflow orchestration, MCP, governed tool execution, evaluation, human-in-the-loop
Distributed systems	Byzantine fault tolerance, intrusion tolerance, real-time systems, event-driven execution, durable state, idempotency, recovery, resilient networking
Security	Sandboxing, network restrictions, enterprise authentication, OAuth2, API keys, mTLS, credential lifecycle, permission scoping, audit
Languages & systems	Python, Go, TypeScript, C, Java, SQL/PostgreSQL, REST/OpenAPI, Prometheus, Grafana, ClickHouse, Git, Linux
Machine learning	PyTorch, TensorFlow, Hugging Face, NumPy, Pandas, intrusion detection, time-series and regression models
Research tools	Mininet, Apache Spark, HP LoadRunner, HP Performance Center
Operating systems	Linux and UNIX environments including CentOS, Ubuntu, Kali, and AlmaLinux; Windows
Development tools	Git, Vim, Visual Studio Code

PRIOR INDUSTRY EXPERIENCE

Aktrix Technologies — **Software Engineer**, Hyderabad, India 2013–2016

- Led and expanded a team of nine performance engineers, overseeing test strategy, resource coordination, delivery risk, and training across client projects.
- Developed an automated performance-monitoring and analysis platform for multi-cloud environments, expanding data-mining and machine-learning analysis while reducing maintenance cost and root-cause-analysis time.
- Built and maintained data pipelines for extracting and cleaning cloud-monitoring and log data; performed feature engineering, model training, and production performance monitoring.
- Developed load and resource-utilization prediction using ARIMA and SARIMA time-series models, decision trees, random forests, XGBoost, regression methods, and deep-learning models.
- Improved latency by approximately 2–5× across projects while reducing root-cause-analysis effort and maintenance costs.

Deloitte — **Performance Engineering Analyst**, Hyderabad, India 2011–2012

- Led performance testing across seven projects and developed approximately 40 performance-test plans for geo-distributed systems.
- Evaluated application performance and network traffic, built monitoring-data pipelines, and analyzed server, network, and resource-utilization data to identify performance bottlenecks.
- Partnered with product owners and application, database, and infrastructure teams to diagnose bottlenecks and optimize transaction latency against service-level objectives.
- Received the Applause Award for performance optimization and root-cause analysis.

TEACHING EXPERIENCE

Machine Learning, Johns Hopkins University — Teaching Assistant Fall 2024
Advanced Distributed Systems, Johns Hopkins University — Teaching Assistant Spring 2022
Distributed Systems, Johns Hopkins University — Special Help Fall 2021
Software for Resilient Communities, Johns Hopkins University — Special Help Spring 2021
Intermediate Programming, Johns Hopkins University — Special Help Fall 2020
Distributed Systems, Johns Hopkins University — Teaching Assistant Fall 2019
Introduction to Artificial Intelligence, Duke University — Teaching Assistant Spring 2019

SELECTED ACADEMIC PROJECTS

Fault-Tolerant Distributed Key-Value Store

Engineered a sharded, replicated key-value store using the Raft consensus protocol to study consistency, availability, network partitions, and data-distribution trade-offs.

Fair Decision-Making with Privacy-Protected Data

Investigated the interaction between fairness and differential privacy, evaluating privacy mechanisms and developing a fairness metric to characterize accuracy and equity trade-offs.

NLP Model Development and Evaluation

Developed and evaluated text-classification and generation models, including fine-tuning BERT-family models with Hugging Face Transformers.

Network and Systems Security

Conducted hands-on security analysis of networked systems and evaluated attack surfaces including authentication, denial-of-service, rogue access points, and man-in-the-middle attacks.

Fat-Tree Performance Evaluation

Built and evaluated Fat-Tree and conventional tree topologies in Mininet, studying the effects of workload placement, scheduling, topology, service-level objectives, and bandwidth utilization.

Haptic Aid for the Visually Challenged

Led development of a computational navigation prototype for indoor use, combining ultrasonic sensing, real-time microcontroller processing, and haptic actuators to help visually impaired users perceive obstacles and navigate more independently.

Full publications, open-source systems, and research materials: b-sahiti.github.io